



Task 5: Your Team's Grid

Quick Start OCPS: Smart Power AI Design Contest · Team Formation

Enter your team name: Late People

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Part 1: Assign Team Roles

Each team member is tasked with genuine technical ownership over a domain of the contest. All four roles contribute to completing objectives throughout every week. Roles defines your primary focus and what you are chiefly responsible for delivering.

Instructions: As a group, read the role descriptions and decide who takes on each role. Write each team member's first and last name in the final column.

Role	Key Responsibilities	Ownership	Member Name
Scrum Master	- Keeps the team moving and removes blockers - Runs the daily standup check-in at the start of each work session - Tracks progress against weekly deliverables and contest deadlines - Leads all communication with the team mentor - Asks 'what is blocking you?' — not 'did you finish?'	- Week 1 Day 1 reflection - Mentor meeting agendas - Final Presentation coordination	Tag
Data Engineer	- Owns the data pipeline — loading, cleaning, and feature engineering - Ensures the correct dataset is prepared for each notebook - Leads the Week 2 deep data exploration work - Builds and maintains the maximum power table - Knows the dataset cold — is the team's go-to on data questions	- all_users_energy.csv data pipeline - hospital_energy_weather.csv pipeline - EDA notebook outputs	Tag
ML Engineer	- Owns the model code across all three notebooks - Leads regression, classification, and optimization modeling decisions - Tunes model parameters and interprets outputs - Explains what the model is actually doing and why - Connects model results to the contest's smart city design brief	- Regression notebook - Classification notebook - Optimization algorithm	Tag
Solutions Architect	- Owns the big-picture design — how the three models form a coherent AI system - Leads the Week 4 design proposal and presentation narrative - Authors the written reflections connecting technical work to real-world stakes - Ensures the team can explain why they made the design choices they did - Presents the final design to the panel of judges	- Week 4 design proposal - Final presentation slides - Written reflections	Grace / Lana

Note: Every team member participates in every activity. Roles define primary ownership, not exclusive participation. The Scrum Master ensures no one is working in isolation.



Part 2: Accessing your team's channel on MS Teams

Instructions: Communicate with your teammates, teacher sponsor, and the contest administrators to ensure that you and your teammates have access to their unique team channel in the Contest Team on Microsoft Teams.

Part 3: Team Data Profile

Instructions: Communicate with your team to agree upon a final **Team Data Profile**, which you created using the **Drawing with Data Notebook**.

Take a **screenshot** of the **Team Data Profile** printed from running the cell.

Paste the screenshot here:



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TEAM DATA PROFILE

Smart City AI Contest – Week 1

Team Name: Late people

Date Completed: 5/18/2026

— DATASET DIMENSIONS —

Total rows: 15,480

Total columns: 14

Date range: 2016-03-26 to 2017-12-31

Time resolution: hourly (hourly / 15-min / daily)

Number of buildings: 4

— MISSING DATA —

Building with most missing data: Ileana (11.7 %)

Building with least missing data: Estela (10.7 %)

Are gaps random or clustered? Random except for Ileana

Our plan for handling missing data: Dont count

— ENERGY RANGES —

Modesto typical range: 329.82 - 568.18 kWh

Justin typical range: 658.13 - 912.31 kWh

Ileana typical range: 26.00 - 56.00 kWh

Estela typical range: 307.00 - 465.57 kWh

— TIME PATTERNS —

Peak hour of day (which building, what hour): Justin, 15

Weekday vs weekend difference (describe): Consumption lowered on weekend

Seasonal pattern visible? Yes, more during warmer months

— RELATIONSHIPS —

Strongest correlation with temperature (building + r value): Justin, 0.682

Weakest correlation with temperature (building + r value): Ileana, 0.185

Most important non-temperature feature: Hour

— DATA CONCERNS —

One data quality issue we found: Large amounts of missing data ~10%

How we plan to handle it: Skip timestamps with missing energy data, as 0 would lower the average creating an incorrect result

— TEAM HYPOTHESIS —

We predict the strongest feature in our regression model will be:

Hour of day

Because:

Has the highest correlation with all buildings, air temperture has a high correlation with Justin, but not other buildings

We will check this hypothesis against our actual model results at the end of the regression notebook.



Part 4: Planning Your Grid

Recap

Reflect upon the exercises you completed using the Java-to-Python and Drawing with Data Notebooks, in which you looked at two different data sets.

- From the **all_users_energy.csv**, you created data frames to select and plot samples of the energy usage of multiple different user types. You processed the energy data to calculate and plot the power usage of selected users, and to calculate and plot the *total* power usage of a theoretical grid containing those users.
- From the **hospital_energy_weather.csv**, you created plots of only a single user type—hospitals—but you accessed more variables/parameters from the data frame so to create different *types* of plots that helped shape a detailed data story.

As you proceed through the contest, your work as AI engineers will involve your team probing more data for a larger number of users, enabling you to inform your design of a smart city grid system with a robust story of the data.

Propose a 10-User Grid

Instructions: Based on what your team learned through Tasks 01–04, propose a grid of 10 important power user types that will make up the grid in your smart city. You can use building types from the **all_users_energy.csv** (the column headings of your initial data frame df) for inspiration.

1. Power Restoration Priorities: Fill out the table below to summarize an initial power restoration plan for your community's new smart grid. Create a numbered list prioritizing the user types you identified, with "Rank 1" being the most important. This should be informed by the background reading in Task 01. *Accurately propose* how many users are in each of your user types will make up your smart city (e.g., 5 Restaurants, 1 Hospital). You may need to do some googling.

User Rank	User Type	User Count
1.	Hospital	1
2.	Fire Station	4
3.	Police Station	3
4.	Clinic	6
5.	Grocery	10
6.	Gas Station	20
7.	Warehouse	8
8.	Bus Station	1
9.	K-12 School	5
10.	College	1

2. Explain why your team decided on these priorities. Were the most essential users prioritized? Or were other factors considered? Write a paragraph using complete sentences.

> The most essential users were prioritized; I primarily used FEMA community lifeline rankings to determine priority. Tier one (Life-sustaining), then tier two (Human needs). This includes schools as



shelters and grocery stores for food distribution and finally tier 3 (economic). This includes basically anything else including bus stations, college and warehouses.



3. Explain any limitations you have in establishing priorities qualitatively. List initial questions you think should be answered by the data in order for an AI model to help establish better priorities.
 - > Establishing priorities qualitatively instead of quantitatively presents the problem of not knowing what actually needs power. While something like a fire station is obviously a high priority, it could probably function with very little to no power or be self-sustaining in the event of an emergency power outage. This is the first question that needs to be answered, what actually needs the most power in an emergency.



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Part 5: Submit Task 05

Instructions:

- Save your **team's shared version** of this document as a **PDF**.
- Name the PDF: **Task 05 late people.pdf**.
- Scrum Master, submit this task via the **Contest Submission Survey**:
https://ufl.qualtrics.com/jfe/form/SV_6KIQfJDpn2lbh7U